Smart Air Quality Forecasting, Health Risk Assessment & Pollution Alert System Using Machine Learning

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Abstract—Fine particulate matter (PM2.5) is a major air-pollution concern in Indian cities. This paper presents a machine-learning system that predicts PM2.5 concentrations from co-measured pollutants and month, maps predictions to project-defined health-risk categories, and issues pollution alerts with precautionary advice. Daily records from six Indian cities—Bengaluru, Chennai, Delhi, Hyderabad, Kolkata, and Mumbai—were cleaned by removing PM2.5 values above a fixed threshold of approximately 300 µg/m³, resulting in 9,114 records. Exploratory analysis identifies Delhi as the most polluted city in the dataset, a seasonal pattern with winter peaks and monsoon troughs, and a strong correlation between PM2.5 and PM10 (r = 0.85). Five regression models were evaluated using a common test split and the metrics R², MAE, and RMSE. Random Forest achieved the best performance (R² = 0.8256, MAE = 11.622, RMSE = 20.3827). A one-sample t-test against a project-selected threshold of 50 rejected the null hypothesis (t = 16.3425, p < 0.05). The system distinguishes predicted PM2.5 concentrations and project-defined categories from the official Central Pollution Control Board (CPCB) Air Quality Index (AQI). It does not calculate the official AQI and is intended for educational and decision-support purposes.

***Keywords*—air quality, PM2.5 prediction, machine learning, random forest, health-risk categories, pollution alert, exploratory data analysis**

I. Introduction

A. Background

Air pollution is a major public-health problem. The World Health Organization (WHO) identifies particulate matter (PM2.5 and PM10), ozone (O3), nitrogen dioxide (NO2), sulphur dioxide (SO2) and carbon monoxide (CO) as the principal pollutants and has tightened its guideline values to reflect evidence of harm at low concentrations [1]. Fine particles below 2.5 µm penetrate deep into the lungs and are linked to respiratory and cardiovascular disease. In India, dense traffic, industrial emissions, biomass burning and winter meteorology push particulate levels in several cities far above recommended limits. The Central Pollution Control Board (CPCB) publishes the National Air Quality Index (AQI) to communicate air quality in simple categories [2], but an index describes present or past measurements; it does not forecast conditions or translate a prediction into personal advice.

Machine-learning (ML) regression can learn the non-linear relationship between co-measured pollutants, season and PM2.5 [3]. Coupling the prediction with a health-risk category and a short alert makes it actionable. The contributions of this work are:

- a preprocessing pipeline that removes extreme PM2.5 values and derives month, season, category and pollution-score features;
- an exploratory study of six cities covering spatial, seasonal and correlation patterns;
- a comparison of five regression models on one shared test split using R², MAE, MSE and RMSE;
- a one-sample t-test of mean PM2.5 against a project-selected threshold; and
- an interactive console module that returns predicted PM2.5, a category, a risk level, an alert and precautions.

B. Terminology and Scope

Three quantities are kept separate throughout (Table I). Predicted PM2.5 is the regression output in µg/m³. The PM2.5 category (Good, Satisfactory, Moderate, Poor) and the health-risk level are project-defined labels assigned from the predicted value by fixed rules in the script; they are educational aids, not regulatory determinations. The official CPCB AQI [2] is a different index with its own categories and breakpoints, computed by CPCB from measured pollutant concentrations. This system neither computes nor reports it, and the project category names must not be read as official AQI categories. The study covers six cities and one target pollutant (PM2.5), and assumes that relationships learned from past days hold for the days being estimated.

TABLE I
QUANTITIES REPORTED BY THE SYSTEM

Quantity	Origin	Status
Predicted PM2.5	Random Forest output (µg/m³)	Model estimate
PM2.5 category	Project rule on predicted value	Project-defined
Health-risk level	Project rule on predicted value	Project-defined
Alert and advice	Fixed project messages	Project-defined
Official CPCB AQI	CPCB, from measured data [2]	Not computed here

II. Problem Statement and Objectives

Public air-quality dashboards share four weaknesses: they report current rather than predicted values; they rarely translate a category into personal precautions; raw sensor data contain spikes that distort statistics and model training; and published studies often report a single algorithm, so readers cannot judge whether a simpler model would suffice. This project therefore (i) cleans and explores multi-city daily data, (ii) compares five regressors on an identical split, (iii) tests mean PM2.5 against a project-selected threshold, and (iv) provides an interactive module that

returns predicted PM2.5, a project-defined category and risk level, an alert and recommendations.

III. Related Work

Méndez et al. survey ML algorithms for air-quality forecasting, organised by algorithm family, target pollutant, input variables and forecasting horizon [3]. Performance in such studies depends on the pollutant, the location, the data quality and the features, which motivates comparing several models under one protocol rather than choosing one in advance. Tree-based ensembles are widely used for tabular environmental data because they capture non-linear interactions without feature scaling; a Random Forest averages many randomised trees grown on bootstrap samples [4]. Linear and low-degree polynomial regression remain transparent baselines. Official portals such as the CPCB AQI service report measured values and index categories [2]. This project differs by attaching a project-defined category, risk level and advice to a predicted PM2.5, and by combining a model comparison with a statistical test and an interactive module.

IV. Methodology and Models

A. Workflow

Fig. 1 shows the pipeline. Daily city records are loaded, cleaned and enriched with features; exploratory analysis characterises the data; five regression models are trained on the same training rows and scored on the same held-out test rows; and the best model drives the prediction, alert and advice module. The pipeline is a single Python script using pandas, NumPy, scikit-learn, SciPy, Matplotlib and Seaborn, run from Visual Studio Code in a PowerShell terminal.

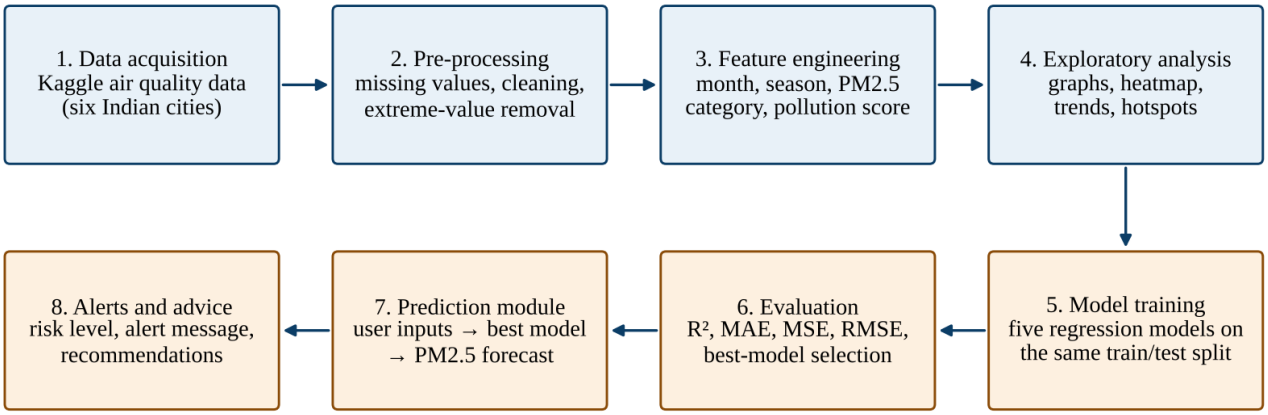


Fig. 1. Proposed system architecture.

B. Models and Evaluation

PM2.5 prediction is a supervised regression problem: the feature vector (PM10, CO, NO2, SO2, O3, month) is mapped to the target PM2.5. Multiple linear regression is the baseline; polynomial regression (degree 2) adds squares and pairwise products; a decision tree splits the feature space with threshold rules; Random Forest averages many decorrelated trees [4]; and gradient boosting adds trees sequentially to correct residual errors. No exhaustive hyper-parameter tuning was performed, so the comparison is a baseline across model families. Performance is measured on the test split with

$$R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2 \quad (1)$$

$$MAE = (1/n)\,\Sigma\,|y_i - \hat{y}_i| \quad (2)$$

$$RMSE = \sqrt{[(1/n)\,\Sigma\,(y_i - \hat{y}_i)^2]} \quad (3)$$

where $MSE = RMSE^2$. Because all results come from a single split, the ranking is stated for this split only; cross-validation would be required to claim general superiority.

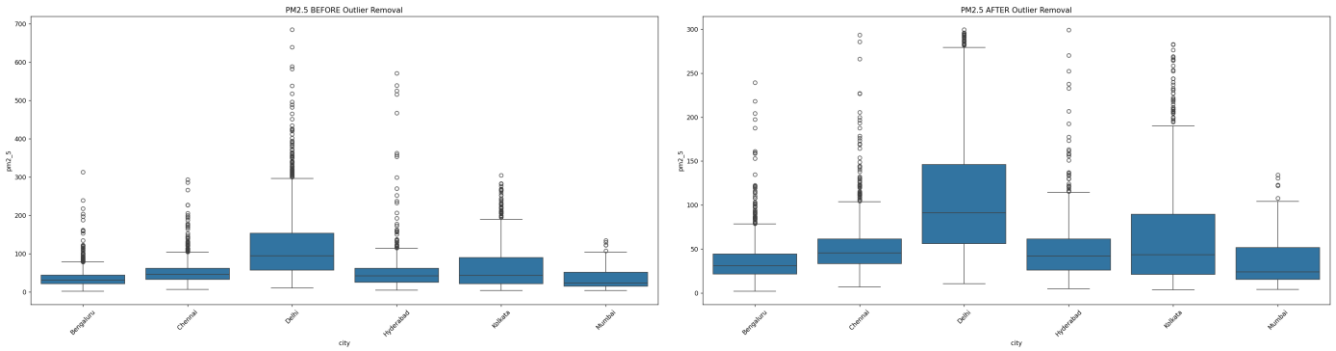
V. Dataset and Preprocessing

A. Dataset

The data follow the city-day format of the Kaggle dataset Air Quality Data in India [5]. Records for Bengaluru, Chennai, Delhi, Hyderabad, Kolkata and Mumbai were used, one record per city per day, starting in 2015. After cleaning, the data contain 9114 rows and 10 columns. The model inputs are PM10, CO, NO2, SO2, O3 and month; the target is PM2.5. Values are in the units of the dataset (PM2.5 and PM10 in µg/m³).

B. Cleaning and Feature Engineering

Column names were standardised, dates parsed, missing values handled before modelling and records sorted chronologically. Derived columns are month, season (Winter, Summer, Monsoon, Post-Monsoon), a project-defined PM2.5 category and a composite pollution score per city used to rank hotspots. Extreme values were removed with a fixed upper cut-off of about 300 µg/m³ (maximum retained: 299.85 µg/m³). Fig. 2(a) shows isolated readings of several hundred µg/m³ (Delhi above 600, Hyderabad above 500) that are far from the bulk of the data and likely reflect sensor spikes or recording errors; Fig. 2(b) shows the result. Winter episodes below the cut-off, notably in Delhi and Kolkata, were deliberately retained because they are the events an alert system must detect. A single global cut-off also removes any genuine severe day above 300 µg/m³, so the most extreme episodes are under-represented.



(a) Before removal (b) After removal

Fig. 2. PM2.5 by city before and after outlier removal.

C. Descriptive Statistics

Table II gives the PM2.5 statistics after cleaning. The mean (58.29) exceeds the median (44.48), indicating a right-skewed distribution in which a minority of heavily polluted days raises the mean. Table III lists the project-defined category counts: 56.2% of records are Good and 30.8% Satisfactory, whereas Moderate and Poor account for 10.0% and 3.0% (271 records). The categories are strongly imbalanced (about 19:1 for Good to Poor), so a few high values dominate MSE and RMSE. These categories are not CPCB AQI categories, which include further classes (Very Poor and Severe) that are not used here.

TABLE II
DESCRIPTIVE STATISTICS OF PM2.5 AFTER CLEANING

Statistic	Value
Count	9114
Mean	58.2945
Standard deviation	48.4536
Minimum	1.7200
25th percentile	27.0100
Median (50%)	44.4800
75th percentile	69.5000
Maximum	299.8500

PM2.5 in µg/m³; values above about 300 µg/m³ removed.

TABLE III
PROJECT-DEFINED PM2.5 CATEGORY COUNTS

Category	Records	Share (%)
Good	5122	56.2
Satisfactory	2809	30.8
Moderate	912	10.0
Poor	271	3.0
Total	9114	100.0

Project-defined labels, not official CPCB AQI categories.

VI. EXPLORATORY DATA ANALYSIS

A. Seasonal Pattern

PM2.5 follows a pronounced annual cycle (Fig. 3): the monthly mean decreases from approximately 94 µg/m³ in January to a minimum of 30 µg/m³ in August, before rising to approximately 92 µg/m³ in December. Season-wise averages (Table IV) support this pattern, with Winter being the most polluted season (88.21 µg/m³) and Monsoon the cleanest (34.34 µg/m³), representing a ratio of approximately 2.6. The observed seasonal variation may be associated with monsoon rainfall, which can remove airborne particles, and winter atmospheric conditions that may limit pollutant dispersion. However, meteorological variables were not included in the dataset, so these explanations remain interpretations rather than tested causal findings.

The seasonal pattern also indicates that air pollution levels vary substantially throughout the year and that a single annual average may not adequately represent exposure conditions. Including month as a model input allows the regression models to capture recurring temporal variation in PM2.5 concentrations. Nevertheless, month alone cannot represent short-term weather changes or unusual pollution episodes. Future work could incorporate rainfall, temperature, wind speed, and boundary-layer conditions to improve seasonal and event-level predictions.

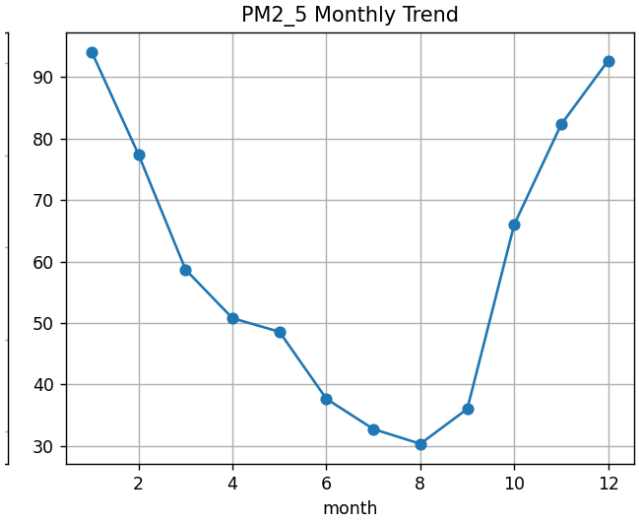


Fig. 3. Monthly mean PM2.5 (PM10 and CO panels omitted).

TABLE IV
SEASON-WISE MEAN PM2.5

Season	Mean PM2.5
Winter	88.2054
Post-Monsoon	73.9262
Summer	52.4849
Monsoon	34.3402

PM2.5 in µg/m³.

B. Correlation

PM2.5 is most strongly correlated with PM10 (r = 0.85), followed by NO2 (0.65), SO2 (0.46) and O3 (0.38); CO shows only a weak correlation (0.15) (Fig. 4). PM10 and NO2 are therefore expected to be the leading predictors,

although CO may still contribute through interactions with other variables.

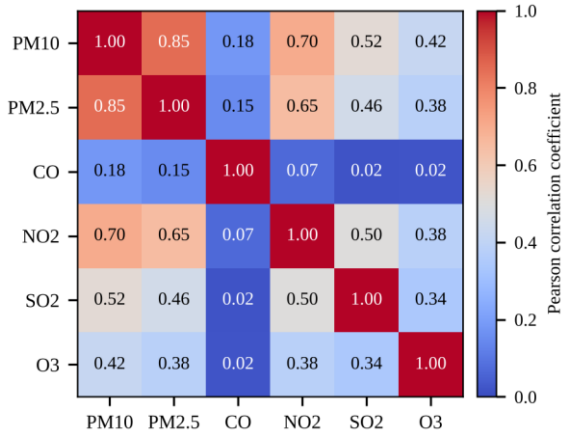


Fig. 4. Correlation heatmap of air pollutants.

C. Hotspots and Extreme Days

The composite pollution score ranks Delhi first (136.11), about 1.8 times Kolkata (74.61); Chennai, Bengaluru, Mumbai and Hyderabad are closely grouped (Table V). Four of the five highest PM2.5 records in the cleaned data belong to Delhi (Table VI), and the most polluted day is 11 December 2019 in Delhi (PM2.5 = 299.85 µg/m³, project category Poor). The second-highest value (299.25) is a Hyderabad record from 21 August 2015, an unusual monsoon-season reading that merits verification as a possible data artefact. Together with the large between-city differences, which are not encoded as an input feature, this suggests that adding the city as a feature is a promising improvement.

TABLE V
POLLUTION HOTSPOT RANKING

Rank	City	Pollution score
1	Delhi	136.1065
2	Kolkata	74.6081
3	Chennai	62.1531
4	Bengaluru	61.9088
5	Mumbai	60.3484
6	Hyderabad	56.9870

Score computed per city by the project script.

TABLE VI
FIVE HIGHEST PM2.5 RECORDS AFTER CLEANING

Date	City	PM2.5
2019-12-11	Delhi	299.85
2015-08-21	Hyderabad	299.25
2016-10-27	Delhi	296.27
2016-11-11	Delhi	295.88
2016-01-01	Delhi	295.71

PM2.5 in µg/m³; values are capped by the outlier filter.

VII. RESULTS AND DISCUSSION

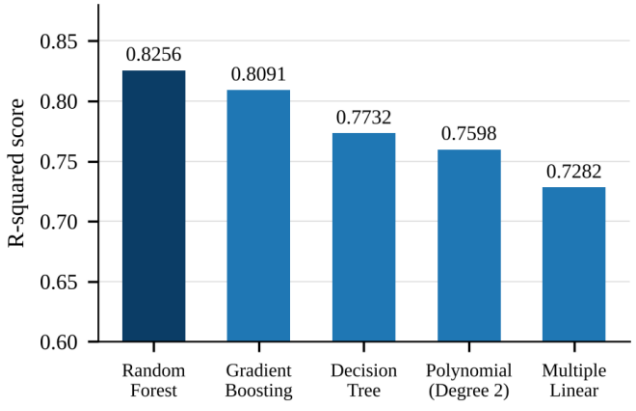
A. Model Comparison

Table VII reports the metrics on the shared test split, as printed by the script. Random Forest ranks first on all four metrics, followed by Gradient Boosting, Decision Tree, Polynomial Regression and Multiple Linear Regression (Fig. 5).

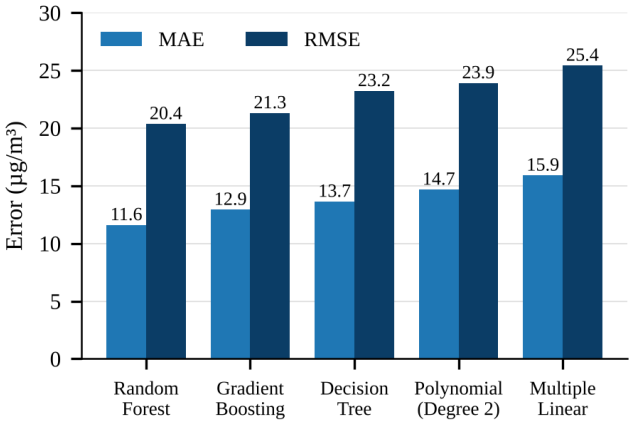
TABLE VII
FIVE-MODEL PERFORMANCE COMPARISON

Model	R^2	MAE	MSE	RMSE
Random Forest	0.8256	11.6220	415.4562	20.3827
Gradient Boosting	0.8091	12.9466	454.7457	21.3248
Decision Tree	0.7732	13.6528	540.2565	23.2434
Polynomial (Deg. 2)	0.7598	14.6972	572.3043	23.9229
Multiple Linear	0.7282	15.9180	647.4334	25.4447

MAE and RMSE in µg/m³ (PM2.5 units).



(a) R^2 by model



(b) MAE and RMSE by model

Fig. 5. Comparison of the five regression models on the shared test split.

B. Discussion

Random Forest explains about 83% of the PM2.5 variance on the test split. Its MAE (11.622) is about 20% and its RMSE (20.3827) about 35% of the mean PM2.5 (58.29). Relative to the linear baseline, MAE is about 27% lower, RMSE about 20% lower and R^2 higher by 0.0974. Both tree ensembles outperform the linear and polynomial models, which indicates non-linear structure, and Random Forest improves on the single decision tree by 0.0524 in R^2. The gap to Gradient Boosting is only 0.0165 and may change with tuning or another split, so that ordering is provisional. RMSE is 1.6 to 1.75 times MAE for every model, which points to occasional large misses, plausibly on pollution peaks; residuals were not analysed by city or season. A mean-value predictor would have an RMSE close

to the standard deviation of PM2.5 (48.45), so even the weakest model gains substantially (Random Forest: about 58% lower RMSE). The remaining unexplained variance (about 17%) reflects influences outside the feature set, such as meteorology and city, so the output should be read as an estimate with a typical error of roughly 12 µg/m³.

C. Statistical Test

A one-sample t-test examined whether mean PM2.5 exceeds a project-selected threshold of 50 µg/m³ (H0: mean = 50; H1: mean > 50) using

$$t = (\bar{x} - \mu 0) / (s / \sqrt{n}) \quad (4)$$

With $\bar{x} = 58.2945$, $s = 48.4536$ and $n = 9114$, $t = 16.3425$ and the one-tailed p-value is 1.70×10^{-59}, so H0 is rejected at $\alpha = 0.05$. The threshold is a project choice, not an official AQI or regulatory limit. Consecutive days are autocorrelated, so the effective sample size is smaller than n and the p-value is strong but approximate evidence (Table VIII).

TABLE VIII
ONE-SAMPLE T-TEST SUMMARY

Quantity	Value
Threshold	50.0
t-statistic	16.3425
One-tailed p-value	1.70e-59
Alpha	0.05
Decision	Reject H0

Threshold is project-selected, not an official AQI value.

VIII. PREDICTION SYSTEM AND HEALTH-RISK ALERTS

A. Module and Worked Example

The best model (Random Forest) is kept in memory and used by a console module. The user enters PM10 (0–500), CO (0–5), NO2 (0–200), SO2 (0–100), O3 (0–200) and month (1–12); the prompts state these permitted ranges and remind the user to enter values in units consistent with the dataset, because the model has learned relations between numbers, not physical units. The module then prints the selected model, the predicted PM2.5, the project-defined category, the health-risk level, an alert and recommendations (Fig. 6).

For PM10 = 120, CO = 2, NO2 = 100, SO2 = 20, O3 = 60 and month = 11, the Random Forest predicts PM2.5 = 57.1, which the project rules label Satisfactory with Medium Risk (Table IX). The alert printed is "Moderate PM2.5 alert." This wording is a project message; it is neither the category name nor an official CPCB category. The two recommendations printed are to consider reducing prolonged strenuous outdoor activity if sensitive and to check local official air-quality guidance. The module provides a simple way to demonstrate the model's predictive capability without requiring users to interact directly with the training code. Displaying the predicted concentration alongside a category and alert makes the output easier to understand. This example illustrates how pollutant measurements and month are used to generate an estimated PM2.5 value. However, the prediction is not a real-time air-quality measurement. Its reliability depends on the accuracy of the entered values and the patterns represented in the training dataset. Therefore, users should consult official air-quality monitoring sources before making health-related decisions.

```
===== PREDICT PM2.5 FROM USER INPUT =====
Prediction uses the best-performing model from this comparison.
Enter pollutant measurements in units consistent with the dataset.
Enter PM10 (0-500): 120
Enter CO (0-5): 2
Enter NO2 (0-200): 100
Enter SO2 (0-100): 20
Enter O3 (0-200): 60
Enter Month (1-12): 11

===== PREDICTION RESULT =====
Selected Model: Random Forest Regressor
Predicted PM2.5: 57.1
PM2.5 Category: Satisfactory
Health Risk: Medium Risk

===== AIR QUALITY ALERT =====
Moderate PM2.5 alert.

===== HEALTH RECOMMENDATIONS =====
- Consider reducing prolonged strenuous outdoor activity if sensitive.
- Check local official air-quality guidance.
```

Fig. 6. Console session of the prediction and alert module.

TABLE IX
OUTPUT OF THE WORKED EXAMPLE (FIG. 6)

Output item	Value	Basis
Selected model	Random Forest	Best on test split
Predicted PM2.5	57.1	Model estimate
PM2.5 category	Satisfactory	Project-defined
Health risk	Medium Risk	Project-defined
Alert text	Moderate PM2.5 alert	Project message
Official CPCB AQI	Not computed	CPCB only [2]

B. Risk and Alert Design

Categories, risk levels and alert texts are fixed project rules intended for education and decision support. The advice is conditional ("if sensitive") because children, older adults and people with respiratory or cardiac disease are affected first [1], and every alert refers the user to official guidance [2]. With a typical test error of about 12 µg/m³, a prediction near a category boundary may fall in the neighbouring category, so the numeric value is shown with the label. The tool is not a medical device and does not replace official advisories.

IX. LIMITATIONS AND FUTURE SCOPE

- **Single split:** The ranking comes from one random split; neighbouring days may fall on both sides, so scores may be optimistic. Cross-validation and confidence intervals are not reported.
- **Limited inputs:** No meteorological, city or lagged variables are used; the model estimates PM2.5 from co-measured pollutants and month, so it is not a multi-day forecast.
- **Data:** Six cities with unequal numbers of valid records, daily means, and a fixed 300 µg/m³ cut-off that also removes genuine extreme episodes.
- **Project-defined labels:** Categories, risk levels and alerts are rule-based, not personalised and not aligned with CPCB AQI breakpoints; the interface is a console only.

Future work will add weather, city and lagged features for a true next-day forecast; k-fold or time-series cross-validation with hyper-parameter tuning; feature-importance

or SHAP explanations; residual analysis by city and season; alignment of categories with official CPCB AQI breakpoints; and a web or mobile interface with live data.

X. Conclusion

This paper presented a machine-learning system for PM2.5 estimation, project-defined health-risk categorisation and alerting for six Indian cities. After cleaning to 9114 records, Delhi emerged as the dominant hotspot (score 136.11, about 1.8 times Kolkata), Winter as the most polluted season (88.21 µg/m³; Monsoon 34.34 µg/m³), and PM10 (0.85) and NO2 (0.65) as the pollutants most correlated with PM2.5. Among five models compared on one test split, Random Forest was most accurate (R^2 = 0.8256, MAE = 11.622, RMSE = 20.3827), and a one-sample t-test (t = 16.3425) showed that mean PM2.5 exceeds the project threshold of 50 µg/m³. Predicted PM2.5, project-defined categories and the official CPCB AQI remain distinct: the system supplies the first two and defers to official sources for the third. With meteorological data, rigorous validation and a graphical interface, the approach could grow into a practical early-warning tool.

Acknowledgment

The author thanks the open data community for making city-level air quality records available and the faculty for guidance on the project.

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